

Document Image Forgery Detection Based on Deep Learning Models

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Abstract—With the improvement of the communication speed and the popularization of the Internet, images have become the most common information medium in life. At the same time, the adverse effects of forged images in the media, credit investigation, finance and academic fields are becoming more and more significant. Therefore, in recent years, the research on forged image identification algorithms has been active worldwide. Image forgery has different classification methods. According to whether the forgery uses deep learning methods, it can be divided into deep forged images and traditional forged images. It can also be divided into ordinary image forged and document image forged according to whether the image is a text image. Different forgery methods will leave different forgery traces in the image, corresponding to different forgery identification methods. Aiming at document forgery images, this paper proposes a forgery detection algorithm based on deep learning and fusion of error level analysis (ELA) information. Compared with the previous forgery identification algorithms, the algorithm in this paper can not only identify whether the document image is forged, but can also locate the forged text area. The algorithm proposed in this paper supports the detection of document image forgery generated by cutting, copying, erasing and deep learning methods. The detection algorithm of this paper participated in the fifth forgery detection competition of Ali Tianchi and won the 32nd place among 1470 participating teams.

Keywords—Document Image Forgery Identification; Deep Learning; ELA Image Processing

I. INTRODUCTION

Compared with written expressions, images are more popular with the public, and therefore they are widely used as information media. In recent years, with the popularization of image editing software, the increasing intelligence of image editing algorithms, and the rise of deep forgery methods, the number of forged images on the Internet has gradually increased, and the quality of forgeries has also been greatly improved. Many forged images cannot be reliably identified by the human eye, and the negative effects they bring cannot be underestimated: fake news, internet rumors, financial fraud and even academic fraud [1]. For example, only in

biology publications, in 40 scientific journals from 1995 to 2004, 3.8% of 20,621 papers contained images of intentional forgery [2]. In December 2017, a user named "deepfakes" posted on the Reddit social networking site a fake video that mapped the faces of actresses to pornographic performers [3]. It triggered a wave of forgery of images and videos with the help of generative adversarial nets (GAN) and other deep learning methods, which are still in continuous development, causing great harm to the people's right of portrait and reputation. Due to the emergence of the corona virus disease 2019 (COVID-19), many work scenarios have changed from offline to online. Uploading document images for certificate verification in e-commerce activities has become a common mode. However, the content of document images can be forged by image editing tools or deep learning algorithms. It is of great significance to verify the authenticity of these certificate document images.

Image forgery technology and authentication technology have formed a confrontational relationship, and both are constantly evolving and progressing. For image forgery, according to its forgery method, it can be divided into the traditional method of using image editing software and the deep forgery method of using deep learning algorithm. Among them, the traditional methods can be divided into four categories: cutting, copying and moving, erasing and image enhancement [4]. Specifically, the cutting method is to stitch together different pictures; the copy method is to copy and move an area in the same picture; the elimination method is to eliminate an area in the image and use other areas pixels filling the eliminated area; image enhancement method is a collective term for a series of image processing methods, including sharpening, blurring, mosaic and so on. The deep forgery method uses deep learning methods, such as autoencoders or generative adversarial networks [5][6]. Input an image to be modified into the model, and the model will output a forged image. The entire process does not require human involvement. Some software open to the public makes deep forgery simpler and more popular, such as ZAO and FaceApp. Deep forged images are mostly

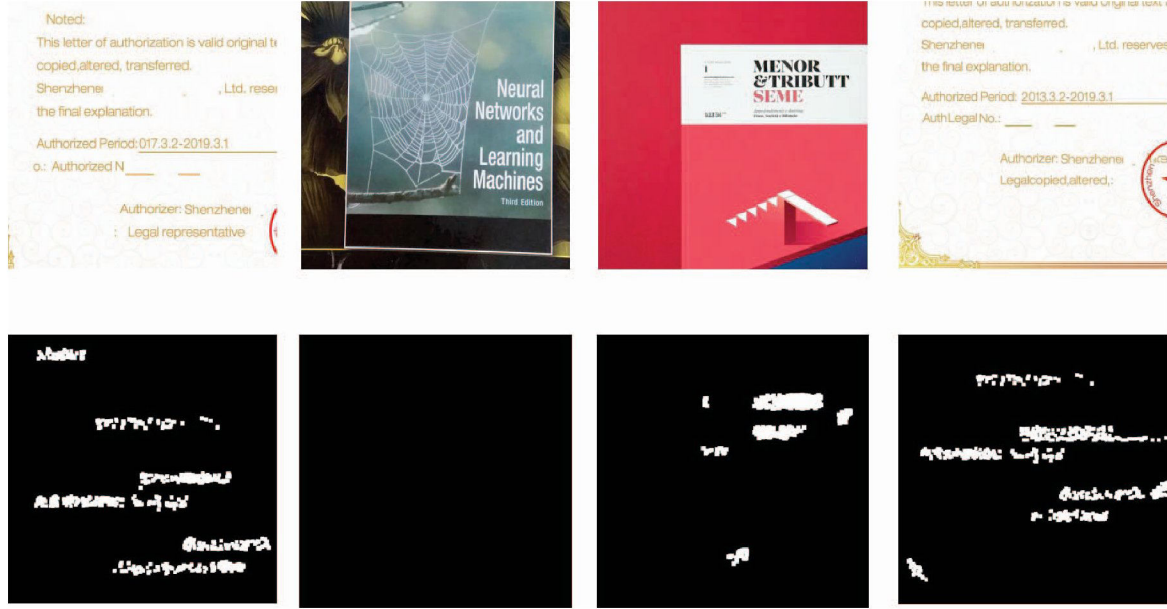


Figure 1. Some images in our dataset. The upper row shows forged images, and the lower row shows corresponding masks. The white area in masks represents that the same area in the image has been forged.

aimed at the face area in the image, and can be divided into four categories according to their forged methods: complete generation, face replacement, attribute modification, and expression replacement. Complete generation: generate completely non-existent face images through deep learning methods, usually based on GAN technology, such as StyleGAN [7] proposed by T. Karras, S. Laine, this technology can generate high-resolution portraits of people, if used in fake resumes, it would be very misleading. Face replacement: Replace the face of a person in an image with a face in another image, usually using the technology of FaceSwap or DeepFakes. This technology can bring some more possibilities to the film production industry. At the same time, there is also the possibility of being used by criminals in the field of pornography and fraud. Attribute modification: One character's face contains some attributes, such as skin color, pupil color, hairstyle, eyebrow shape, etc. The attribute modification is to modify these attributes, such as modifying the skin texture, changing the age profile of the character, or adding glasses to the character. It is also implemented based on GAN, such as StarGAN [8] proposed by Y. Choi, M. Choi, M. Kim, J. Ha, S. Kim. Expression replacement: Modify the expression of the character in the image to the expression of the reference character. The most famous of this type of forgery are the

Face2Face method and the NeuralTextures method [9][10].

For document image forgery, most of the traditional methods are used, as described above, including forgery methods such as cutting, copying, moving, erasing, and image enhancement. However, in recent years, as deep learning has become popular in the field of forgery, there also existed ways to forge document images through deep learning methods, such as SRNet [11] proposed by Wu et al, STEFANN [12] proposed by Roy et al. This year, Lin Zhao, Changsheng Chen and Jiwu Huang published a paper, showing deep learning methods can be used to modify text in document images even with complex backgrounds [13].

II. RELATED METHODS

The identification of forged images is based on a basic assumption: there will be features left by the forgery process in the forged images. These features are not necessarily visual features, and they are not necessarily discernible by the human eye. These clues include JPEG compression features [14], edge consistency features [15], noise patterns [16], color consistency features [17], visual similarity [18], EXIF consistency [19], camera model features [20], image semantic features and hidden layer features left by deep forgery [21], etc.

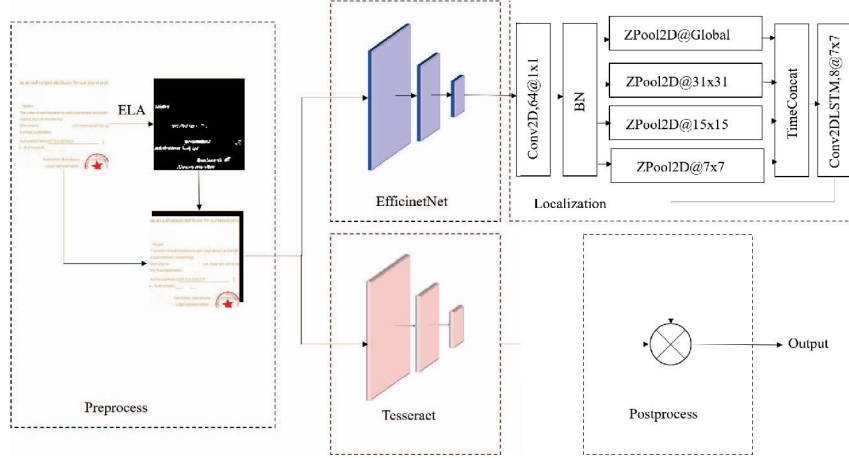


Figure 2. Network architecture.

In terms of specific implementation, the identification technology can be divided into traditional identification methods and deep learning identification methods according to whether deep learning methods are used. In recent years, more and more research work is based on deep learning, and traditional methods such as analysis by means of noise mode [16] or analysis by means of DCT correlation [17] have been seldom used. Research on identification using deep learning methods is very popular. Researchers use different backbone networks to extract the hidden layer information in the image, supplemented by different pre-processing and post-processing, to identify whether the image has been forged [22][23][24]. Most of existing detection algorithms only support classification tasks, that is, they cannot determine where area of a forged image has been forged. There are also algorithms that can support the location of the forged area, such as ManTra-Net [25].

In this paper, we propose a new algorithm for document image forgery identification based on deep learning and fusion of Error level analysis (ELA). It not only supports determining whether the image has been forged, but also supports giving forged areas. Moreover, it can detect forgery by cutting, copying, moving, erasing, enhancing, and deep learning methods, which is of great significance to the forgery identification of document images.

III. NETWORK ARCHITECTURE

As shown in Figure 2, the network proposed in this paper is composed of five parts. The preprocessing part is used to generate error level analysis (ELA) channels and concatenate it with RGB channels. The EfficientNet part is used to extract image features, which is to perform ELA information fusion.

Error level analysis is a traditional forensic method to identify portions of an image with a different level of compression. It could be used to determine whether a picture has been modified. ELA can expose different compression levels, if one image was modified, it would show inconsistent compression levels between different areas of the image, and this information can help us determine whether an image has been forged. Specifically,

during JPEG compression, the picture will be divided into 8×8 independent small blocks, and each small block will be subjected to independent lossy compression. The forged image will be subjected to additional lossy compression in the forged area. Therefore, there will be obvious differences between the forged area and the background area in the ELA image. If there are forged regions in a picture, the compression levels between different regions on the picture are no longer similar, which provides clues for forgery detection. The more times of forgery, the more obvious the difference in ELA image. In the traditional ELA forgery detection method, after generating the ELA image of the image to be detected, it will be handed over to experienced experts to analyze and judge the ELA image. Taking the expert's experience as the standard to judge whether it is forged or not is time-consuming and laborious, and the detection results may also be affected by the personal level of experts, which cannot ensure the objectivity of the results.

In this paper, we do not need ELA experts. We manually calculate the ELA map of the image, and use the ELA map as additional channels to the traditional RGB channels, and then input the model as a whole, so that the model can learn information from both RGB channels and the ELA channels. The ELA channels contain more forgery information when forged images leave few clues in RGB channels, so that when facing some images with inconspicuous visual forgery features, the model can still learn the forged regions in the image through the ELA channels. The backbone network of this part uses EfficientNet [26]. Generally speaking, the way to improve the capability of a convolutional neural network is nothing more than deepening or widening the network, but whether it is deepening or widening, it will greatly increase the computational cost. There is no theoretical calculation formula to refer to how to choose the coefficient of deepening or widening. In this context, EfficientNet uses the technology of Neural Architecture Search to adjust the depth and width of the model, so that EfficientNet can spend lower computing power in image classification tasks to obtain better classification accuracy than large networks such as ResNet-152 [27]. Therefore, here we choose EfficientNet as the backbone network for extracting image features.

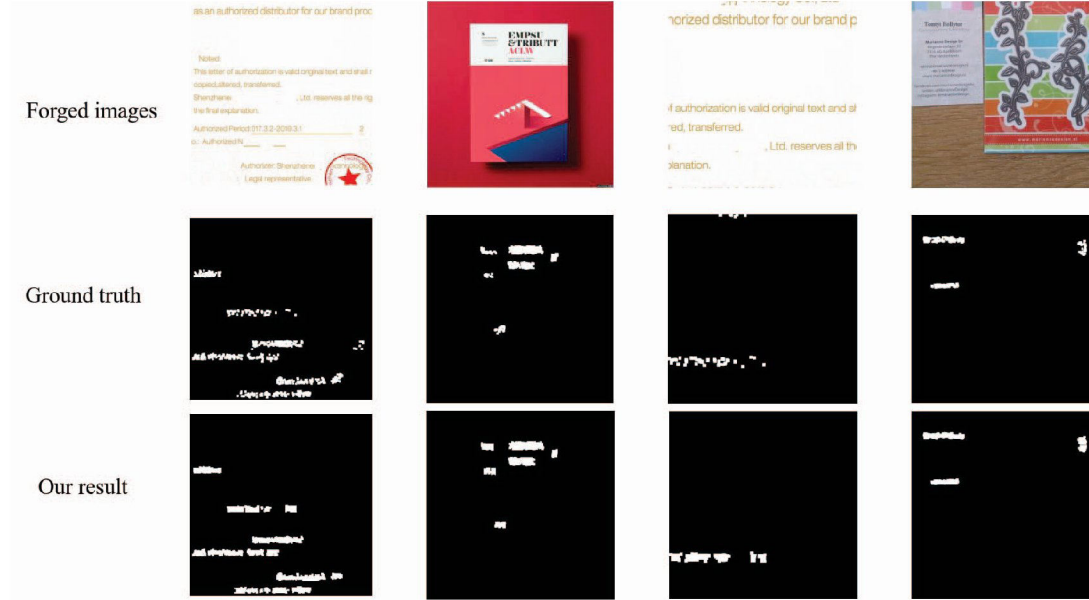


Figure 2. Model result. The upper row shows forged images, the middle row shows corresponding ground truth masks and the last row shows our models output. The white areas in masks represent forged areas.

The function of the location part of the model is to locate the fake areas. This part of the model takes the output of the EfficientNet as input, that is, the input of the location part is the feature of the image, and the output is a Mask image that is the same size as the original image. In this part, we use the architecture of Local Anomaly Detection Network [25]. The value of each pixel is a floating point number between 0 and 1, which represents the probability that the pixel has been forged: 0 represents not forged, and 1 represents forged.

The tesseract part and postprocessing part of the model is responsible for result optimization. For tesseract, the input is the original image, and the output is the coordinates of each text in the image. Tesseract is an open source text recognition tool that supports English and Chinese characters. Finally, the postprocessing part of the model corrects the Mask image output in the localization part according to the text area of the tesseract part to obtain the final result.

The innovations of our model are as follows:

- Our model can not only detect whether the image has been forged, but also give the forged areas.
- The fusion of ELA features and RGB features can make full use of ELA features to improve the detection effect in the face of forged images with unclear visual information.
- In the localization module, pooling kernels of different sizes are used to simulate the characteristics of far and near when human eyes observe the image, so as to fully mine the clues of forged image, and finally better locate the forged area.

IV. EXPERIMENT

A. Dataset

The dataset used in this article is provided by the fifth forgery identification competition hold by Alibaba Tianchi. Figure 1 shows some pictures of the dataset. The

dataset contains a total of one thousand pictures, including photographs and scans of various types of certificates. The forgery methods used in the dataset include cutting, copying and moving, erasing, image enhancement, and a combination of various methods. In addition, all images in the dataset are additionally compressed with JPEG to counter detection.

B. Data Enhancement

For the dataset of Ali Tianchi, we use a variety of methods to enhance the data. There are 1,000 original images in the dataset, which can be considered as a small dataset in situation of deep learning tasks. Data enhancement methods can play a role in expanding the dataset. In addition, the generalization of the model can also be improved. Specifically, we perform random cropping, mirroring, and rotation operations on each image to expand the dataset to 5,000 images.

C. Model Realization

In this paper, we use pytorch to implement the proposed model, we have discussed our model structure in section 2. We choose ReLU as activation function, choose SGD as the optimizer. In addition, the momentum is set to 0.9, the initial learning rate is set to 0.001, and decreased by multiplying 0.1. As to evaluation metrics, we choose AUC.

TABLE I. MODELS' ACCURACY TEST FOR ALL TYPES OF FORGERY

Algorithm	AUC
EOI1 [31]	48.1%
CFA1 [28]	49.2%
J-LSTM [29]	74.6%
RGB-N [30]	81.5%
Ours	82.3%

TABLE II. TABLE 2. MODELS' ACCURACY TEST FOR COPYING AND REMOVAL FORGERY, USING AUC AS METRIC.

Algorithm	Copying	Removal
EOI [31]	48.2%	49.1%
CFAI [28]	48.6%	49.4%
J-LSTM [29]	72.3%	76.6%
RGB-N [30]	80.6%	82.0%
Ours	81.8%	82.7%

D. Experimental Result

Figure 3 shows detection results in Ali Tianchi dataset. For mask images, black areas represent not being forged, while white areas represent being forged. We can see that our model can exactly expose most forged areas, only tiny forged areas missed in our model's output. Besides.

Table 1 shows accuracy comparison between the model in this paper and other existing models. The results show that on the dataset mixed with all forgery methods, our model beats other models and achieved best result.

Further, we tested the two most common forgery types copying and removal separately. The test results are shown in Table 2. The results show that our model is better than the existing baseline models for forgery detection of copying type or removal type. In conclusion, our model can effectively detect forged images.

V. CONCLUSION

In this paper, we propose a document image forgery detection model based on deep learning and fusion of error level analysis (ELA) information, which can not only detect whether the image has been forged, but also give the forged areas. The fusion of ELA features and RGB features can make full use of ELA features to improve the detection effect in the face of forged images with unclear visual information. It supports the detection of forgery by cutting, copying, moving, erasing, and deep learning methods. On the public dataset provided by the fifth forgery detection competition held by Ali Tianchi, it has achieved a comparable result to that of the international best algorithms. Besides, the algorithm in this paper also has some shortcomings, one of which is that our algorithm can locate the forgery area but cannot distinguish the forgery category. This weakness requires further research.

ACKNOWLEDGMENTS

This research work was partially supported by the National Natural Science Foundation of China (Grant No.61801448, No. U20B2063).

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